
AWS Config – Overview for Solutions Architects

What is AWS Config?

AWS Config is a **service that continuously monitors, records, and evaluates the configuration of your AWS resources**. It helps ensure compliance, governance, and auditability across your AWS environment.

Key Use Cases

Use Case	Example
Compliance auditing	Ensure all EC2 instances are of approved types
Security & governance	Detect unencrypted S3 buckets
Change tracking	Who changed a security group rule and when?
Resource inventory	List all RDS DB instances across regions
Automated remediation (via Systems Manager)	Auto-restrict public access if an S3 bucket is misconfigured

Core Components

1. Configuration Recorder

- Captures **configuration changes** to supported AWS resources
- You must specify the **resource types** to track

2. Delivery Channel

- Sends **snapshots** and **change logs** to an **S3 bucket**
- Also streams changes to **Amazon SNS**

3. Configuration Items (CIs)

- JSON snapshots of your resource state at a specific point in time
- Includes metadata, relationships, and current settings

4. Config Rules

- Evaluate resources against **desired configurations**
- **AWS Managed Rules** (prebuilt) and **Custom Rules** (Lambda-based)

Examples:

- `s3-bucket-server-side-encryption-enabled`
 - `ec2-instance-managed-by-systems-manager`
 - Custom rule to check naming conventions
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How It Works

1. Resource changes (e.g., EC2 tags, security groups)
2. Config Recorder captures and sends to S3
3. Config evaluates changes against rules
4. Violations are flagged (non-compliant)
5. Optional: Remediate via AWS Systems Manager Automation

💡 Architectural Considerations

Factor	Recommendation
Central governance	Use aggregators across accounts & regions with AWS Organizations
Storage	Use a dedicated S3 bucket with lifecycle policies
Notification	Set up SNS topics to notify SecOps teams when violations occur
Remediation	Use SSM Automation documents to fix issues automatically
Compliance reporting	Integrate with AWS Security Hub or third-party SIEMs

🧠 Exam Tips (AWS SAA-C03)

- AWS Config is **region-specific**, but can **aggregate data** across regions/accounts
 - You can **view historical configurations and relationships**
 - Config **does not prevent changes**, it only detects/report them
 - To **automatically fix issues**, use **Config + Systems Manager**
 - AWS Config can evaluate **both current state and drift** from baseline
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💰 Pricing Summary

Component	Pricing Model
Recorded resources	Charged per configuration item per region

Rules evaluations Charged per rule evaluation

Stored snapshots S3 pricing applies

 Example: Enforce Encryption on S3 Buckets

1. **Enable AWS Config**
 2. Select rule: `s3-bucket-server-side-encryption-enabled`
 3. Choose remediation action: **Lambda function or SSM automation**
 4. Any non-encrypted S3 bucket is flagged as **Non-Compliant**
 5. Notification sent to SNS
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